

Recurrences III (Dividing)

Dividing-function recurrences by substitution.

Watch these Abdul Bari videos

- > 2.3.1 Recurrence Relation Dividing Function $T(n)=T(n/2)+1$ #1
- > 2.3.2 Recurrence Relation Dividing [$T(n)=T(n/2)+ n$]. #2
- > 2.3.3 Recurrence Relation [$T(n)= 2T(n/2) +n$] #3

Key concepts

- $T(n)=T(n/2)+1 \rightarrow O(\log n)$.
- $T(n)=2T(n/2)+n \rightarrow O(n \log n)$ (merge sort).
- Draw the recursion tree to sum work.

Key idea

$2T(n/2)+n$ has $\log n$ levels each doing $O(n)$ work $\rightarrow O(n \log n)$.

Tasks (1.5h)

- ☐ Watch the 3 videos
- ☐ Solve $T(n/2)+1$
- ☐ Solve $2T(n/2)+n$

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0